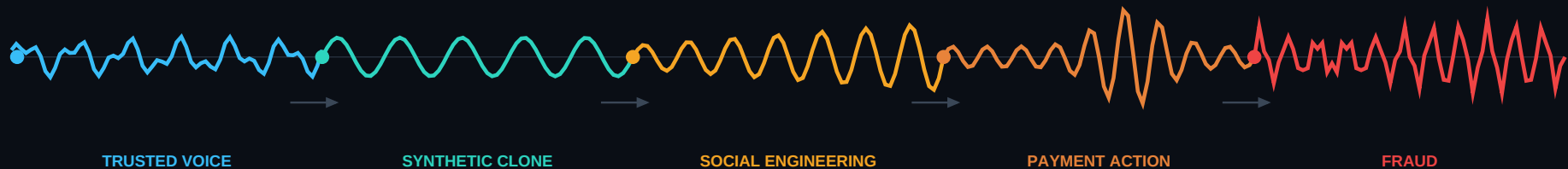


The voice you trust can now be synthetic.

AI voice cloning turns a trusted financial conversation into an attack surface.



A cloned voice — built from **3–10 s** of public audio, in an Indian language, over an ordinary phone line — manufactures urgency and turns a trusted call into a payment, a beneficiary change, or a disclosed OTP.

VoxShield — Real-Time Voice-Clone Fraud Detection for Safer Banking & Payment Workflows

A privacy-conscious detection layer that scores suspicious voice interactions in real time and returns an **evidence-backed risk signal** for human or policy review. **It does not accuse, block, or deny a transaction from a model score alone.**

5.9%

EER · held-out in-the-wild

42→82%

Indic deepfake recall

<1 s

verdict latency / GPU

On-prem

air-gapped · no egress

Team DigiSeva · Track 01 · Working prototype (TRL-5). Project evaluation; not independently validated by the hackathon organisers.

A trusted voice, weaponised inside a live financial interaction

The **primary user** is a bank / NBFC fraud & trust-and-safety operations analyst — and behind them, every Indian phone customer targeted by a cloned voice.

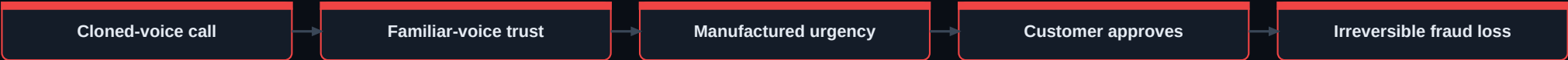
The harmful moment (BEFORE)

- Attacker clones a trusted person's voice; calls during a live banking / payment interaction.
- Customer trusts the familiar voice → social engineering → approves a transfer, changes a beneficiary, or discloses an OTP.
- Fraud completes in seconds. Recovery is slow and rarely successful.

Who sees the signal — and what they may do

Question	Answer
WHO receives it	Authorised fraud-ops analyst
WHAT they see	Risk state + evidence + confidence
WHEN	Within the call's first seconds (streaming)
ACTION allowed	Step-up verify / escalate / review
NOT allowed	Auto-block / accuse from a score alone

The current journey — today, with no defense



The India-specific deployment challenge. Much commercial voice-deepfake detection is cloud/API-oriented and English-centric, which can create deployment and data-governance constraints for sensitive Indian telephony workflows (aligned with DPDP and RBI data-localisation expectations). VoxShield is designed for on-prem, Indian-language, phone-quality audio — and CERT-In has issued deepfake advisories as the threat grows.

Detect the signal before trust becomes loss — detection $\neq$ accusation

VoxShield: detect the signal before trust becomes loss. The product turns a blind judgment call into an evidence-backed, human-governed decision — **detection $\neq$ accusation.**

Product flow (AFTER)

- **Voice** → controlled, on-prem analysis of the live call.
- **Detection** → five detectors produce independent evidence.
- **Evidence + risk state** → LOW / REVIEW / HIGH, explained.
- **Verification** → REVIEW triggers step-up (callback / 2nd channel).
- **Human review** → HIGH escalates to authorised fraud-ops.
- **Safe action** → a person decides; the model never does.

The system communicates “evidence indicates elevated voice-clone risk” — never “this person is a fraudster.”

What the analyst actually sees

VoxShield · Voice Authenticity Evidence

UI EXAMPLE

RISK STATE: HIGH

Review required — evidence indicates elevated voice-clone risk

91%

✓ Multiple detector agreement (4 / 5)

✓ Synthetic spectral characteristics

✓ Abnormal F0 / prosody dynamics

✓ Vocoder-like phase regularity

✓ Indic-TTS synthesis signature

A risk signal — not proof of fraud.

→ Step-up verification / authorised fraud review

Illustrative analyst console — the 91% is an example risk signal for a single call, not a benchmark result.

Six pillars — a safe financial-risk workflow, not a generic detector

Not “another deepfake detector.” VoxShield converts voice-clone detection into a **safe financial-risk workflow** — six pillars competitors and generic builds leave open.

Telephony-first

Trained for 8 kHz G.711 phone audio, not studio clips — measured codec cost.

Indic-aware

XLS-R-53 backbone + Indic routing across 10 Indian languages.

Streaming

3 s window / 1 s hop → a verdict within a call's first seconds.

Multi-detector fusion

Five detector families + learned meta-fusion; no single point of evasion.

Explainable evidence

Per-signal reason codes, not a bare score — legible to a fraud analyst.

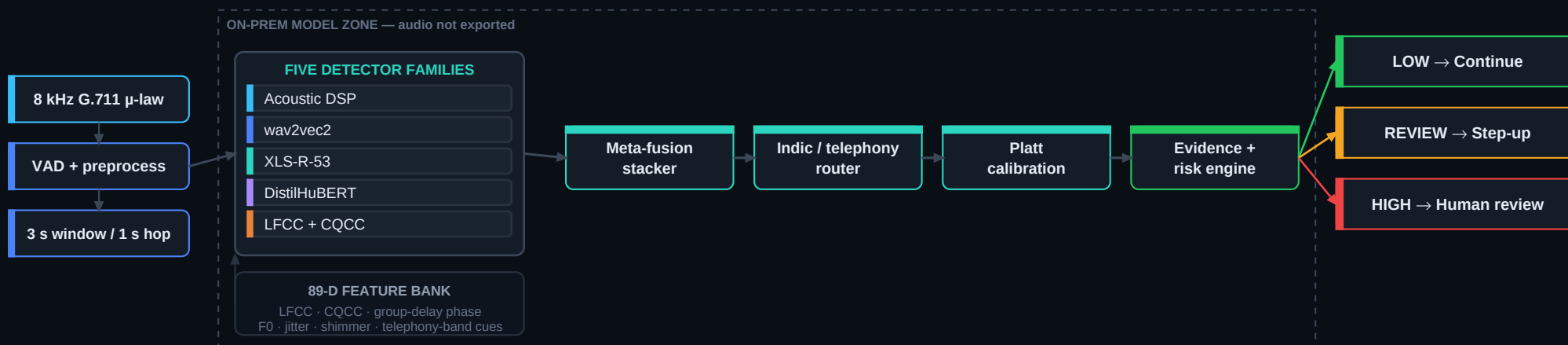
Safe escalation

LOW/REVIEW/HIGH, human-in-the-loop, never auto-blocks. Fails safe.

Phone audio + Indic language + Multi-model evidence + Financial workflow + Human governance = VOXSHIELD

From 8 kHz telephony to a calibrated, explainable risk signal

Five complementary detectors over a shared 89-D feature bank → **learned meta-fusion** (not a naive average) → Indic/telephony routing → calibration → a calibrated risk state. Multiple signals, one governed decision.



Under 8 kHz telephony, information above the Nyquist limit is not available; the system uses cues present within the available telephony band. Routing priority is Indic → telephony → general, fail-safe. **Detection ≠ accusation.**

Measured, scoped, and honestly bounded — no hero numbers

Measured on project corpora under disjoint splits, reported with scope. In-corpus and cross-dataset figures are never conflated.

5.9%

Fusion EER · held-out in-the-wild (800 clips)

42 → 82%

Indic deepfake recall (MMS-TTS)

11.7% → 6.3%

Genuine-Indic false positives (channel-aware)

Methodology

- Speaker-disjoint; cross-generator holdout (train on some cloners, test a held-out cloner); cross-dataset OOD (MLAAD); per-language & codec-degraded breakdowns.
- Fusion beats the best single detector (16.0% EER) and a naive average (19.1%) — the honesty check.
- ROC-AUC 0.983; calibration ECE 0.044. G.711 codec cost $\approx$ 1.6 pts cross-dataset.

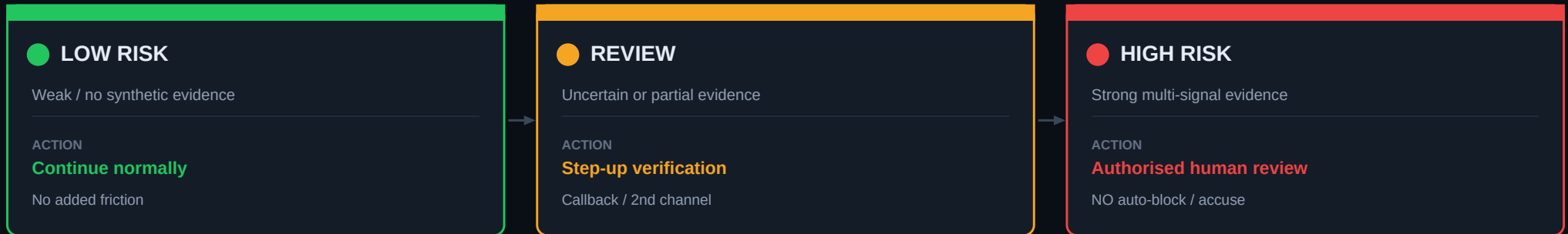
Project evaluation; not independently validated by the hackathon organisers. Axes never truncated to exaggerate.

Caveats (stated, not hidden)

- EER is not accuracy; Indic recall is not overall recall.
- In-the-wild phone figures are in-domain-optimistic; the robust cross-dataset finding is the $\approx$ 1.6-pt codec cost.
- Multi-cloner Indic cross-generator numbers are in progress.

When AI is wrong, the product must fail safely

When AI is wrong, the product must fail safely. A wrong high-impact action on a genuine customer is itself a harm — so the model's output is **evidence for a decision, never the decision**.



Trust guarantees

- **No automatic blocking or accusation** from a score alone.
- **Privacy by design:** raw audio scored in the authorised boundary; least-data logging.
- **Auditability:** tamper-evident log of every verdict and action.
- **Explainability:** reason codes, not a bare score.
- **Fairness:** genuine-customer false-alarm rate tracked per language.
- **Misuse resistance:** access-controlled evidence; no fraudster look-up tool.

A risk-event source in the fraud stack — people and policy decide

VoxShield is a **risk-event source** inside the existing fraud stack — it emits signals; the bank's people and policies act on them.



Risk-event payload (outputs)

risk_state · evidence[] · confidence · model_version · language · latency_ms · timestamp · audit_id — a structured event, easy to route into an existing case-management queue.

Latency & operational assumptions

- <1 s measured verdict latency on a single GPU; horizontal throughput scales via inference workers.
- Passive scoring first — no customer-facing change; step-up hooks added later.
- Alert routing tuned to REVIEW/HIGH only, to avoid analyst alert fatigue.

One narrow, provable slice: live call → evidence → safe human action

A deliberately narrow finale MVP: **a live Indian-language call, scored on-screen in real time, with evidence and a safe human-review action** — on synthetic / consented audio.

MVP scope (must-have)

- Streaming ingestion → 5 detectors → fusion → LOW/REVIEW/HIGH.
- Evidence explanation + safe escalation + privacy boundary.
- Benchmark dashboard on synthetic / public / consented test data.

Nice-to-have

Indic expansion · authorised-context mock · adversarial-test dashboard · model-update stub. **Do not overbuild.**

Workstreams (owner · hours · demo proof)

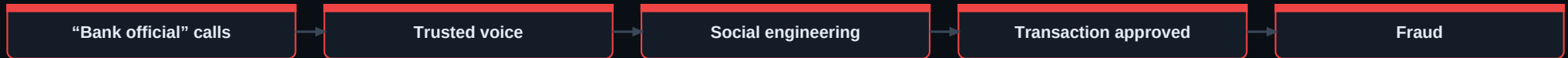
Workstream	Owner	Hrs	Demo proof
Ingestion + VAD + streaming	Member 1	0–10	Live waveform in
Detectors + fusion + routing	Member 2	10–26	5 scores + verdict
Risk engine + explainability	Member 3	18–26	Evidence panel
Workflow + escalation + UI	Member 4	26–40	Review action
Security / privacy / logging	M1+M3	34–40	Audit log + boundary
Eval + failure cases + demo	All	40–48	Benchmark readout

Dependencies: phone-audio capture path (Twilio/SIP or replayed WAVs), one GPU, held-out eval clips. Fallback = replayed recordings if live telephony is unavailable.

Make voice a signal — not a blind spot

Make voice a signal — not a blind spot.

BEFORE · today



AFTER · with VoxShield

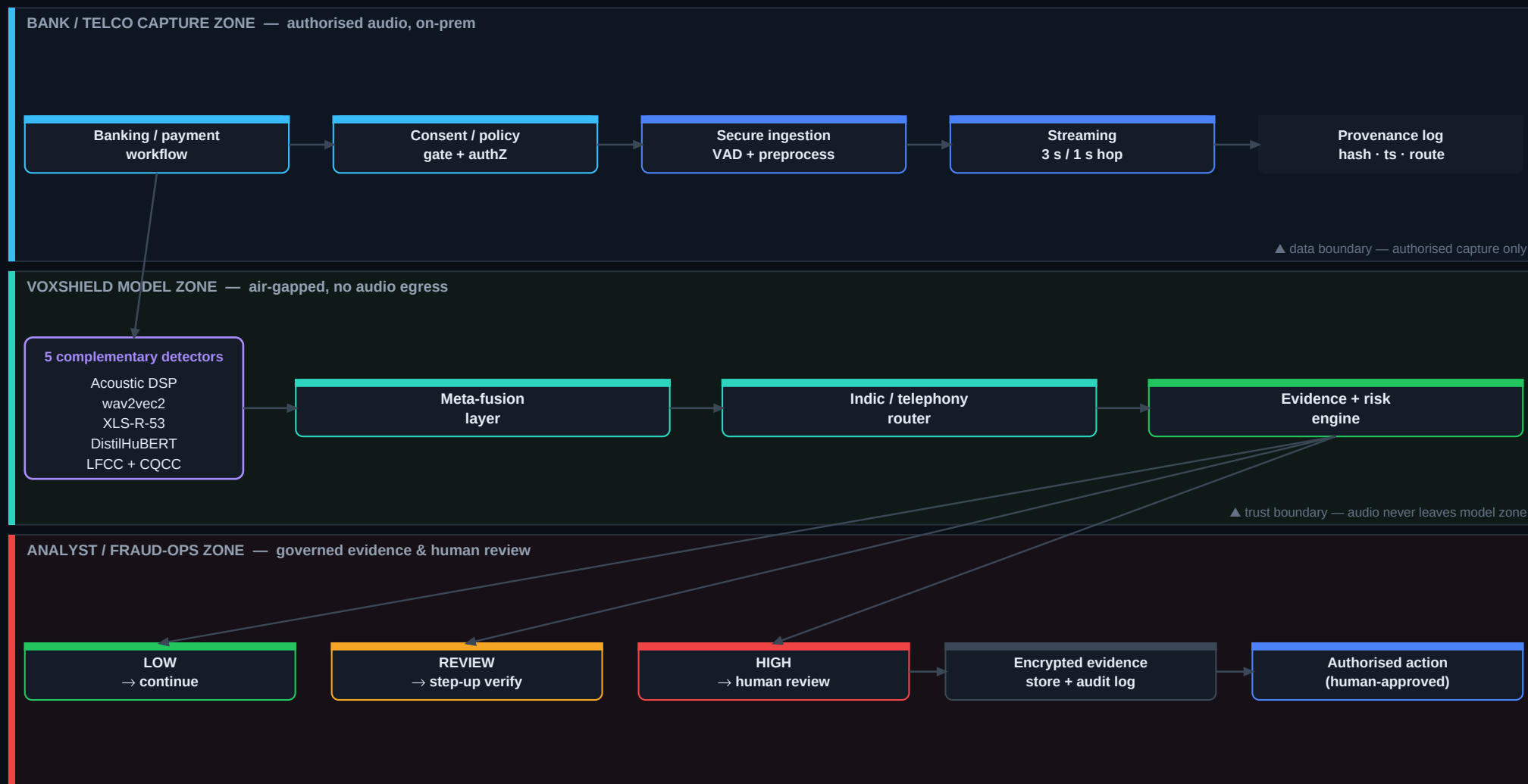


VoxShield does not decide whether a person is a fraudster. It detects whether the voice signal deserves scrutiny — and routes that uncertainty safely.

VoxShield helps financial teams detect suspicious synthetic voices without turning an imperfect model into an irreversible decision.

Capture → model → analyst, with trust & data boundaries

One end-to-end system across three trust zones (read bottom-up). **Audio never leaves the model zone**; only risk states, evidence and tamper-evident records cross into the analyst zone.



Input → preprocess → detectors → fusion → risk → escalation

The decision path from input to escalation — deterministic, explainable, and safe by construction.

**Detection → decision**

- Input (8 kHz call) → VAD + preprocess → 3 s/1 s streaming windows.
- 5 detectors → meta-fusion → Indic/telephony router → Platt calibration.
- Optional **authorised contextual signals** (future integration / prototype mock) adjust risk — never the sole trigger.
- Evidence + risk engine → LOW / REVIEW / HIGH.

Escalation & outputs

- LOW → continue · REVIEW → step-up verification · HIGH → authorised human review.
- No automatic block or accusation at any state.
- Emits a structured risk event (state, evidence, confidence, version, language, latency, audit id).

Authorised contextual signals (caller-identity mismatch, unusual beneficiary, abnormal timing, recent account changes, repeated failed verification) are **future integration points / prototype mocks**. We do not claim access to bank transaction data.

Minimisation, consent, encryption, audit, adversarial mitigation

Data & privacy

- **Data minimisation:** collect only what is required to score the call.
- Raw audio processed in the authorised boundary; **not exported**. Least-data logging (hashes, verdicts, reason codes).
- If audio is retained for an authorised investigation: explicit authorisation, encryption, access control, audit logging, limited retention, deletion policy.
- We do **not** claim “zero data storage” — we claim controlled, minimised, auditable storage.

What we collect / don't

Aspect	Position
Collect	Call audio frames + minimal metadata (under authorisation)
Process where	Inside the bank's authorised boundary, on-prem
Retain	Verdicts/evidence by policy; raw audio only if authorised
Access	Role-based; logged
Don't collect	No external export; no unrelated PII

Security & model protection

- On-prem / air-gapped deployment (DPDP + RBI localisation aligned).
- Encrypted evidence store; role-based access; tamper-evident audit log.
- Secure model serving; rate limiting; hidden/rotated thresholds where appropriate.
- Adversarial evaluation + drift detection + secure update mechanism.

Human review is mandatory for high-impact action

Every high-impact outcome passes through an authorised human. The model narrows attention and supplies evidence; it never executes an irreversible financial action.

Boundaries (honest)

No complete adversarial robustness is claimed. Real noise/music robustness needs Indic fine-tuning of the backbones (roadmap). Contextual signals are mocks until integrated.

Five families · 89-dim feature bank · learned meta-fusion

Five complementary detector families over a shared 89-dim feature bank, combined into one calibrated risk score. Diversity is the anti-evasion property.

Detector	Family / signal	What it catches
Acoustic DSP	Hand-built spectral / prosodic DSP (ours)	Vocoder + phase artifacts; narrowband cues
wav2vec2-base	SSL transformer (deepfake clf)	General synthetic-speech representation
wav2vec2-large-xlsr-53	Multilingual SSL — 53 langs, ~315M	Indic-aware synthetic cues
DistilHuBERT	Distilled SSL (fast, deepfake clf)	Lightweight corroborating signal
LFCC + CQCC head	Trained cepstral logistic head (ours)	Classic anti-spoofing front-end

Feature bank (89-dim) — verified

20 LFCC + 20 CQCC (mean+std = 80) + 9 interpretable scalars (HF ratio/regularity, spectral flatness, group-delay phase, F0 jitter, shimmer, voiced ratio, silence, breath) = **89**. Analysis at 16 kHz; under 8 kHz telephony, information above the Nyquist limit is not available, so the model relies on cues within the telephony band.

Fusion & calibration

A **trained LFCC/CQCC logistic head + Platt calibration**, blended with the neural detectors via Indic/telephony routing and a recall-aware max-vote (a confident fake from any trusted detector is not averaged away) → LOW / REVIEW / HIGH + reason codes. Not a naive average.

Synthetic / public / consented research data only — no confidential data

Only synthetic, public, or consented research datasets — no confidential bank, customer, or platform data. Each dataset is used under its public research-release terms.

Dataset	Purpose	Licence / consent	Usage	Retention
In-the-Wild	Real/spoof eval (headline EER)	Public research release	Evaluation	Eval only
MLAAD	Cross-dataset OOD	Public research release	Evaluation	Eval only
IndicSynth (ACL 2025)	Indic synthetic fakes	Public research release	Train + eval	Working set
IndicVoices (AI4Bharat)	Real Indic speech	Public corpus (AI4Bharat)	Train + eval	Working set
DFADD	English diffusion/spoof fakes	Public research release	Train + eval	Working set
Rural (Bhojpuri)	Real low-resource speech	Public research release	Train + eval	Working set
Self-generated TTS (MMS-TTS...)	Attack diversity	Self-produced (synthetic)	Train + eval	Working set

No confidential bank, customer, or platform data is used. Datasets are used under their public research-release terms; the full per-dataset licence texts are catalogued in the team data catalogue. garystafford is used as a balanced telephony eval set. CodecFake+ and BanglaFake are downloaded but not yet integrated into the training set. Self-generated fakes are produced in-house for attack diversity.

Trained · fine-tuned · pretrained · open-source · proposed

Full disclosure of models, their role, and how we use them.

Model / asset	Role	Source	Our status
wav2vec2-large-xlsr-53 (~315M)	Indic detector backbone	facebook/...-xlsr-53	PRETRAINED → FINE-TUNED BY US
wav2vec2-base deepfake clf	SSL detector (demo ensemble)	MelodyMachine (HF)	OPEN-SOURCE
xlsr-53 deepfake clf	SSL detector (demo ensemble)	Gustking (HF)	OPEN-SOURCE
DistilHuBERT deepfake clf	Lightweight detector (demo)	Om-Parab (HF)	OPEN-SOURCE
Acoustic DSP + LFCC/CQCC	Feature front-ends	Classic DSP	IMPLEMENTED BY US
Fusion head + Platt calib.	Combine detectors → risk	Ours	TRAINED BY US
Silero VAD	Voice activity detection	Open-source	OPEN-SOURCE (unmodified)
Liveness / context layer	Anti-replay / context	—	PROPOSED

AI-assisted writing/layout: this document was drafted with AI assistance and reviewed by the team. All quantitative results are the team's own measured experiments — not AI-fabricated. Libraries: PyTorch, HuggingFace Transformers/Datasets, torchaudio, soundfile, reportlab.

Metrics × conditions × honest status labels

Metrics × conditions, with honest status labels. We separate what is measured from what is a target or not-yet-measured.

Metrics

EER · ROC-AUC · precision · recall · FPR · FNR · detection latency · time-to-warning · robustness (noise, compression) · unseen-generator · per-language Indic.

Status labels

CURRENT measured today · **TARGET** goal · **TBM** to be measured at finale.

Condition	Metric	Status
Clean / in-the-wild	5.9% EER · AUC 0.983	CURRENT
Telephony (G.711)	≈1.6 pt codec cost	CURRENT
Indic (MMS-TTS)	recall 42%→82%	CURRENT
Genuine-Indic fairness	FP 11.7%→6.3%	CURRENT
Noisy / compressed	robustness	TBM
Unseen generators	cross-generator EER	TARGET / TBM
Replay / attack	detection rate	TBM

Project evaluation; not independently validated by the hackathon organisers. We do not fabricate future results.

What happens when the system is wrong — both directions

False positives (harm to genuine users)

Causes: unusual genuine voice, illness, poor call quality, background noise, compression, code-switching, accents, speech impairment, emotional speech, unfamiliar speakers.

Handling: a score over threshold is never “fraud confirmed.” Evidence → risk state → contextual checks → human/policy decision. Genuine false-alarm rate is tracked per language; REVIEW adds verification, not a block.

False negatives (attacker evasion)

Attackers adapt: new TTS/VC engines, replay, short utterances, threshold probing.

Handling: ensemble diverse detector families; evaluate unseen generators; maintain adversarial test sets; monitor drift; update models; combine with authorised context; never present detection as proof of fraud.

Bias: Indic fairness is a first-class metric — an earlier augmentation experiment that regressed genuine-Marathi false positives was caught and reverted. Fairness is measured per language, not assumed.

Adversarial threats × mitigations · raise cost, fail safe

Adversarial threats and mitigations. No complete robustness is claimed — the goal is to raise attacker cost and fail safe.

Threat	Mitigation
Adaptive TTS / voice conversion	Ensemble diversity; unseen-gen eval
Replay attacks	Liveness / anti-replay (proposed)
Compression / noise injection	Telephony-aware training; robustness eval
Short-utterance evasion	Streaming windows; min-evidence gating
Model / threshold probing	Rate limiting; hidden/rotated thresholds

Threat	Mitigation
Evidence theft	Encryption; access control; audit log
Unauthorised access	RBAC; on-prem boundary
Data poisoning	Curated updates; provenance checks
Model extraction	Secure serving; rate limiting
Drift over time	Drift detection; secure update mechanism

Server → on-prem CPU → edge/NPU (quantised, roadmap)

The on-prem constraint is a design driver, not an afterthought — the roadmap targets edge-class inference for cost and air-gapped deployability.

Deployment tiers

- **Server GPU** (today): A100 / consumer RTX; <1 s measured verdict latency; scale via workers.
- **On-prem CPU / small GPU**: DistilHuBERT-led lightweight route.
- **Edge / NPU** (roadmap): quantised INT8 + ONNX for embedded / handset-class hardware.

Optimisation roadmap

- ONNX export + INT8 quantisation of the detector stack.
- Distillation to a compact narrowband model for telephony.
- Optional Snapdragon / NPU acceleration for edge fraud-ops appliances.

Status: PROPOSED / roadmap — not yet benchmarked on edge silicon.

Narrow MVP, hour-by-hour, with a demo proof at every stage

Hour-by-hour engineering plan for the finale MVP — deliberately narrow, with a demo proof at each stage.

Hours	Focus	Deliverable / demo proof
0–4	Environment, data prep, skeleton	Repo + synthetic/consented eval set staged
4–10	Audio ingestion · VAD · streaming	Live waveform → windows on screen
10–18	Detector integration · inference	5 detector scores on a live clip
18–26	Fusion · risk engine · explainability	LOW/REVIEW/HIGH + evidence panel
26–34	Financial workflow · escalation	Review/step-up action wired to a mock queue
34–40	Security · privacy · logging	Data boundary + tamper-evident audit log
40–46	UI · dashboard · live demo	Analyst console + benchmark readout
46–48	Testing · failure cases · rehearsal	Frozen demo + per-language fairness check